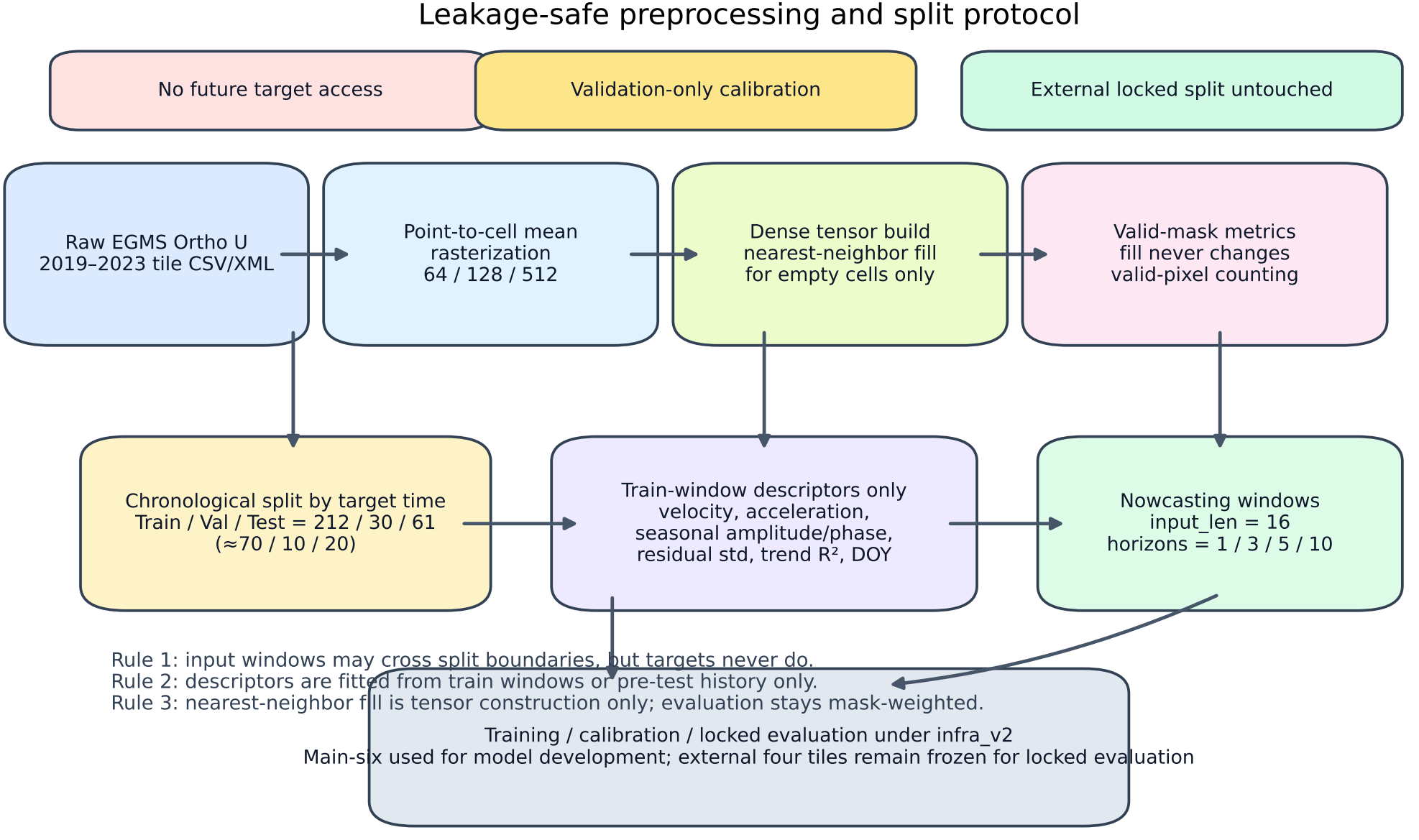
Leakage-safe preprocessing and split protocol
No future target access
Validation-only calibration
External locked split untouched
Raw EGMS Ortho U
2019–2023 tile CSV/XML
Point-to-cell mean
rasterization
64 / 128 / 512
Dense tensor build
nearest-neighbor fill
for empty cells only
Valid-mask metrics
fill never changes
valid-pixel counting
Chronological split by target time
Train / Val / Test = 212 / 30 / 61
(≈70 / 10 / 20)
Train-window descriptors only
velocity, acceleration,
seasonal amplitude/phase,
residual std, trend R², DOY
Nowcasting windows
input_len = 16
horizons = 1 / 3 / 5 / 10
Rule 1: input windows may cross split boundaries, but targets never do.
Rule 2: descriptors are fitted from train windows or pre-test history only.
Rule 3: nearest-neighbor fill is tensor construction only; evaluation stays mask-weighted.
Training / calibration / locked evaluation under infra_v2
Main-six used for model development; external four tiles remain frozen for locked evaluation